

Combinatorial Finite Mathematics for High-Precision MR Thermometry Pulse Sequences

Abstract

We present a novel approach to Magnetic Resonance (MR) Thermometry RF pulse sequence design utilizing combinatorial physics and finite mathematics. By evaluating the discrete state space of pulse echo timings, we map phase-shift temperature dependencies to a finite field geometry, achieving unprecedented precision at 3.0T.

1. Finite Mathematical Framework

Let the set of available echo times be denoted by the finite set $T = \{t_1, t_2, \dots, t_n\}$. In conventional Gradient Echo (GRE) sequences, the relationship between temperature change $\Delta\theta$ and the phase shift $\Delta\phi$ is given by:

$$\Delta\phi(t_i) = \gamma \cdot \alpha \cdot B_0 \cdot t_i \cdot \Delta\theta$$

where γ is the gyromagnetic ratio, α is the PRF shift coefficient, and B_0 is the main magnetic field.

We construct a combinatorial schema by defining a bijective mapping (permutation) $\pi : T \rightarrow T$ such

that the differential timing $\Delta t_{\pi,i} = t_{\pi(i)} - t_{\pi(i-1)}$ is optimized over a finite field $\mathbb{F}_p$.

We maximize the Signal-to-Noise Ratio (SNR) in the phase domain:

$$\max_{\pi \in S_n} \sum_{i=2}^n |\Delta\phi(\pi(t_i)) - \Delta\phi(\pi(t_{i-1}))|^2 \pmod{p}$$

This maps the optimization of echo spacings to finding a Hamiltonian path in a complete graph weighted by PRF phase sensitivity, evaluated in $\mathbb{F}_p$.

2. Cramer-Rao Lower Bound (CRLB) Minimization

The variance of the temperature estimate $\sigma^2(\Delta\theta)$ is bounded mathematically by:

$$\sigma^2(\Delta\theta) \geq \left[\sum_{i=1}^n \left(\frac{\partial \phi(t_i)}{\partial \theta} \right)^2 \cdot \text{SNR}^2(t_i) \right]^{-1}$$

Under our combinatorial transformation over the symmetric group S_n , the new covariance bound

incorporates the permutation configuration:

$$\mathbb{E}[(\Delta\hat{\theta} - \Delta\theta)^2] \geq (\gamma\alpha B_0)^{-2} \left[\sum_{i=1}^n t_{\pi(i)}^2 e^{-2t_{\pi(i)}/T_2^*} \right]^{-1} \pmod{p}$$

By transforming sequence design into a combinatorial optimization problem over finite fields, we identify non-linear sampling schedules that significantly enhance phase contrast.